

claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

```
- Source: `claude-wsl-ubuntu`  
- Kind: `claude`  
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d\subagents\workflows\wf_b0aa0abe-638\agent-a1158fbc4f69f0976.jsonl`  
- SHA-256: `b54caf1054f9d36d1982e29c2ca710fa7c9dccccf4a7fb6b434b2bc442dea653f`  
- Source modified: `2026-06-29T17:08:26+00:00`  
- Imported at: `2026-07-05T16:48:30+00:00`  
- project: `wf_b0aa0abe-638`  
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`
```

Transcript

1. user (2026-06-29T17:05:48.183Z)

Merge these four investigation results into ONE ordered implementation plan for the badminton-highlight-indexer wrap-up. Repo rule: ONE PR per work item, branch off master, no stacking, run `python run\_all\_tests.py` + ruff + mypy + coverage (must not drop), then merge.

Results JSON:

```
[  
  {  
    "key": "issue-398-servedgate",  
    "data": {  
      "summary": "Root cause of #398: served_gate_a.served_handoff_windows builds idx_cfg as a  
plain Python dict and passes that RAW dict directly into the production seams  
seg._enrich_candidate_stats(...) and seg._select_for_handoff(...), bypassing process_video /  
_normalize_config. In production, process_video does idx_cfg = self.config.indexing, which is a  
typed IndexingConfig (because BasePlaySegmenter._normalize_config ran AppConfig(**config),  
coercing the nested dict to typed). So production hits fusion_hybrid.py:135 `fcfg =  
idx_cfg.fusion` on a typed FusionConfig and works; the litmus hits it on a dict and raises  
`AttributeError: 'dict' object has no attribute 'fusion'`. The `idx_cfg_typed: Any = idx_cfg`  
line (served_gate_a.py:85) is only a type-checker silencer ? it stays a dict at runtime. Note:  
_build_runner/_build_native_runner already defensively coerce dict->IndexingConfig, but  
_enrich_candidate_stats and _select_for_handoff (the only two seams the litmus calls) do  
not.\n\nCONFIRMED PRODUCTION PASSES A TYPED CONFIG: trajectory_hybrid.py:190 `idx_cfg =  
self.config.indexing` where self.config is an AppConfig (AppConfig.indexing is `IndexingConfig =  
Field(default_factory=IndexingConfig)`, and IndexingConfig.fusion is `FusionConfig`). Verified  
by reproduction: AppConfig(**{'indexing': <dict>}) yields self.config.indexing typed as  
IndexingConfig and .fusion typed as FusionConfig.\n\nCHOSEN FIX = Option A (preferred): make  
_served_idx_cfg return a typed IndexingConfig. This exercises the real production contract ? the  
seams now receive exactly the IndexingConfig type process_video hands them, and it validates the  
dict against extra='forbid' (catches a future typo'd served key). Verified end-to-end on the  
15-clip golden set with Option A applied in-memory: the whole litmus runs with NO crash and  
reproduces every pinned band.\n\nOption B (make _enrich_candidate_stats / _select_for_handoff  
tolerate dict-or-typed) is REJECTED: it would have to coerce in two seams, weakens the typed  
contract the test exists to exercise, and lets the litmus pass a type production never sees ?  
defeating the litmus's whole purpose ('drives exactly the production methods ? no  
reimplementation').\n\nPINNED BANDS DO NOT NEED RE-PINNING for this fix. Running the now-fixed  
litmus on the existing 15-clip golden set reproduces the 2026-06-22 snapshot within every band:  
served mean_delta_f1 = -0.0233 (band -0.06..0.03 PASS), mean_f1_on/mean_f1_off = 0.9017 (>= 0.5x  
PASS), mean_seg_ratio_on 0.4696 <= off 0.5825 (PASS), every video n_handoff_off>=n_handoff_on  
and num_gt>0 (PASS); PROPOSAL test mean_delta 0.0287 (approx 0.03 +/-0.03 PASS), seg_off 1.364  
(approx 1.36 PASS), seg_on 0.792 (approx 0.79 PASS). So the bands are correct as-pinned; they  
will only need re-measurement when the golden corpus itself changes (which the test docstring  
already calls out), not because of this fix.\n\nCOVERAGE IMPACT: these 5 tests currently SKIP  
everywhere (the bug means they would crash if run; in practice they skip because the default  
manifest output/human_lovo_manifest.json is absent). On the owner box, set  
SERVED_GATE_A_MANIFEST=/home/avidullu/projects/khelsutra-guru/badminton-highlight-
```

indexer/output/p2b_golden/manifest_local.json (exists, 15/15 specs present on disk) with cv2 importable (4.13.0 present) to make all 5 tests RUNNABLE and GREEN. CI still skips cleanly (no golden manifest, cv2 importorskip) ? do NOT make CI depend on golden data.",

"exact_changes": "FIX (Option A) ? backend/eval/served_gate_a.py\n\n1) Add the typed import. At line 33 (after `from backend.eval.metrics import Interval, Score, score`), insert:\nfrom backend.config.models import IndexingConfig\n\n2) Change _served_idx_cfg to RETURN A TYPED IndexingConfig (not a dict).\n\nBefore (served_gate_a.py:43-57):\n\n def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:\n \"\"\"...returns config dict...\"\"\"\n return {\n \"fusion\": {\n \"substrate\": \"wasb\", \n \"min_crossings\": min_crossings, \n \"min_players\": 0, \n \"cue_flags\": {}, \n \"wasb\": {\"velocity_threshold\": PROD_VELOCITY_THRESH}, \n \"dead_time_merge_gap\": PROD_MERGE_GAP, \n \"min_action_window_duration\": PROD_MIN_WINDOW, \n }\n }\n\nAfter:\n\n def _served_idx_cfg(min_crossings: int) -> IndexingConfig:\n \"\"\"The TYPED `IndexingConfig` the served FusionSegmenter seems read ? exactly the\n type `process\_video` passes (`self.config.indexing`). Mirrors config.json's\n `indexing.fusion` block, Gate A at `min\_crossings`, person cue OFF (no GPU/torch).\n Built typed (not a raw dict) so the seams get the real production contract and\n `extra='forbid'` rejects a future typo'd served key.\"\"\"\n return\n IndexingConfig(\n fusion={\n \"substrate\": \"wasb\", \n \"min_crossings\": min_crossings, \n \"min_players\": 0, \n \"cue_flags\": {}, \n wasb={\"velocity_threshold\":\n PROD_VELOCITY_THRESH}, \n dead_time_merge_gap=PROD_MERGE_GAP, \n min_action_window_duration=PROD_MIN_WINDOW, \n })\n\n3) In served_handoff_windows, drop the now-pointless type-silencer alias and use idx_cfg directly.\n\nBefore (served_gate_a.py:79-92):\n\n idx_cfg = _served_idx_cfg(min_crossings)\n seg = FusionSegmenter(None, {\"indexing\": idx_cfg, \"sport\": \"badminton\"})\n # The served per-window stats (incl. net_crossings) via the production enrichment hook.\n idx_cfg_typed: Any = idx_cfg\n # type: ignore[annotation-type-mismatch]\n window_stats = [\n seg._enrich_candidate_stats(\n cw, points, fps, frame_width, video_path=\"\", \n idx_cfg=idx_cfg_typed\n)\n for cw in windows\n]\n keep = seg._select_for_handoff(windows, window_stats, idx_cfg_typed)\n\nAfter:\n\n idx_cfg = _served_idx_cfg(min_crossings)\n seg = FusionSegmenter(None, {\"indexing\": idx_cfg, \"sport\": \"badminton\"})\n # The served per-window stats (incl. net_crossings) via the production enrichment hook ?\n # idx_cfg is a typed IndexingConfig, exactly what process_video hands these seams.\n window_stats = [\n seg._enrich_candidate_stats(\n cw, points, fps, frame_width, video_path=\"\", idx_cfg=idx_cfg\n)\n for cw in windows\n]\n keep = seg._select_for_handoff(windows, window_stats, idx_cfg)\n\nNotes:\n- Passing {\"indexing\": idx_cfg(typed), \"sport\": \"badminton\"} to FusionSegmenter still works: _normalize_config sees a dict, runs AppConfig(**config); pydantic accepts a typed IndexingConfig instance for the `indexing` field. (Optionally pass AppConfig(indexing=idx_cfg, sport=\"badminton\") for full typing, but it is not required for the fix.)\n- The `Dict` import in served_gate_a.py is still used elsewhere (specs: List[Dict[str, Any]]), so leave the typing import line as-is. `Any` is still used (idx_cfg dict, etc.), leave it too.\n\nNO change needed to backend/pipeline/segmenters/fusion_hybrid.py ? line 135 `fcfg = idx\_cfg.fusion` is correct against the typed contract.\n\nRE-MEASURE (owner box) command, only needed if the corpus changes (not for this fix):\n\nSERVED_GATE_A_MANIFEST=/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/output/p2b_golden/manifest_local.json\n\npython -m pytest tests/test_served_gate_a_regression.py -q\n(or `python -m backend.eval.served\_gate\_a --manifest output/p2b\_golden/manifest\_local.json` for the table). Requires cv2 importable (present: 4.13.0).",

"files_touched": [\n "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served_gate_a.py"\n],

"new_or_updated_tests": [\n "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_served_gate_a_regression.py ? no edits required to the assertions; the existing 5 tests (test_served_path_runs_on_all_golden_videos, test_served_gate_a_does_not_materially_regress_f1, test_served_gate_a_does_not_increase_over_segmentation, test_served_uses_production_windowing_operating_point, test_proposal_windowing_gate_a_is_f1_neutral_and_cuts_over_seg) become RUNNABLE+GREEN once the seam crash is fixed and SERVED_GATE_A_MANIFEST points at output/p2b_golden/manifest_local.json.

Optional hardening (not required): add a fast non-golden unit test asserting served_gate_a._served_idx_cfg(2) returns an IndexingConfig whose .fusion.min_crossings==2 and that FusionSegmenter._enrich_candidate_stats accepts it without AttributeError on a tiny synthetic trajectory ? this would catch the regression in CI WITHOUT golden data (the golden-gated tests cannot)."

```

    ],
    "coverage_note": "Before fix: all 5 tests skip everywhere (golden manifest absent in CI; and if a manifest were present they would CRASH on the dict.fusion AttributeError). After fix on owner box (SERVED_GATE_A_MANIFEST=output/p2b_golden/manifest_local.json, cv2 4.13.0 present, 15/15 specs on disk): all 5 run and pass, reproducing the 2026-06-22 pinned bands (served ?F1 -0.0233; f1_on/f1_off 0.9017; seg_on 0.4696 <= seg_off 0.5825; PROPOSAL ?F1 0.0287, seg_off 1.364, seg_on 0.792). CI behaviour unchanged: skips via the _golden_specs() >=6 gate (no manifest) and pytest.importorskip('cv2'); the fix introduces NO golden dependency in CI. Recommend the optional synthetic-trajectory unit test so the dict-vs-typed regression is covered in CI too.",
    "risks_or_unknowns": "Low risk. (1) The fix only changes the eval/litmus helper, not the production fusion_hybrid seam, so production behaviour is byte-identical. (2) extra='forbid' on AppConfig/IndexingConfig accepts the served dict today (verified ? all keys valid: fusion/wasb/dead_time_merge_gap/min_action_window_duration); if a future edit adds an unknown served key it will now fail loudly at _served_idx_cfg construction ? that is the intended stricter contract, but worth knowing it converts a silent-ignore into a hard error. (3) The pinned numbers are a snapshot of THIS 15-clip corpus; they remain valid only while output/p2b_golden/manifest_local.json's clips are unchanged ? the test docstring already flags re-measurement on corpus growth (this is unrelated to the bug fix). (4) Bands were verified against output/p2b_golden/manifest_local.json (the issue's manifest); the test's DEFAULT manifest path is output/human_lovo_manifest.json which is absent on this box, so without the env override the suite still skips ? set SERVED_GATE_A_MANIFEST to run."
  }
},
{
  "key": "issue-397-straydirs",
  "data": {
    "summary": "Root cause confirmed and reproduced. The stray `gcs://`, `gs://`, `local://` directories at the repo root are written by the per-video observability report writer (obsreport), NOT by the storage layer. When a scheme URI (`gcs://?`, `gs://?`, `local://?`) is passed as `video_path` to `emit_indexer_report`, obsreport derives the report directory `\"next to the video\"` via `os.path.dirname(os.path.abspath(video_path))`. `os.path.abspath` collapses the `scheme://` to `scheme:/` and, since it has no leading `./`, treats it as a RELATIVE path, joining it under the cwd (the repo root). Then `os.makedirs` creates `/gcs:/bucket/videos/`, etc.\n\nWHERE the URI leaks in: two callers in backend/main.py pass the RAW request URI instead of the resolved local path:\n- backend/main.py:953 (API path) ? `video_path=req.video_path`\n- backend/main.py:1725 (CLI path) ? `video_path=args.video_path`\n\nThe worker (backend/worker/_main_.py:298) already passes the resolved `local_video` and is NOT affected. backend/api/job_worker.py:78 routes back through main.py, so it is covered by the same fix.\n\nThe `local:/tmp/pytest-of-avidullu/...` tree comes from the test suite: tests/test_api_endpoints.py drives the API path with bucket/local-scheme URIs (test_process_bucket_uri_yields_queryable_video_id [gs://, gcs://], test_process_nonlocal_uri_skips_exists_and_fetches [gs://], test_process_local_scheme_uri_accepted [local://], plus the CLI test_cli_nonlocal_input_fetches_and_keeps_uri which produced the `cli.*` artifacts). These tests do NOT mock emit_indexer_report and obsreport IS installed, so real reports got written to the repo root.\n\nTHE FIX (surgical, 2 lines): pass the already-resolved local path `local_video` (returned by storage.acquire_local_input) instead of the raw URI at both main.py call sites ? exactly what the worker already does. `acquire_local_input` returns the bare stripped path for local:// (so reports land next to the real file) and a scratch tempfile path for gs://gcs://s3:// (so reports land in the temp scratch dir) ? never a scheme-prefixed string, never the repo root. DB provenance is untouched (db.add_video still stores req.video_path/args.video_path as the source URI). `local_video` is in scope at both finally blocks (assigned at main.py:804 before the try at 828; and main.py:1658 before the try at 1662).",
    "exact_changes": "(a) EXACT WRITER + WHY\n- Ultimate writer: /home/avidullu/projects/khelsutra-guru/sports-obsreport/src/obsreport/io.py:48-49\n  if video_path:\n      return os.path.dirname(os.path.abspath(video_path)) or \"./\", name\n  Reproduced: os.path.abspath(\"gcs://bucket/videos/x.mp4\") -> \"<repo_root>/gcs:/bucket/videos/x.mp4\" (abspath collapses \"//\" to \"/\") and, lacking a

```

```

leading \"\", joins under cwd). dirname -> \"<repo_root>/gcs:/bucket/videos\". Then io.py:71-72
`write_text` does os.makedirs(directory, exist_ok=True), creating the scheme-prefixed dir at
the repo root. Same for gs:// and local:// (local:///tmp/... ->
\"<repo_root>/local:/tmp/...\".)\n- The bad value ENTERS at the host caller (this repo): the URI
is passed straight through as video_path.\n    backend/main.py:953 video_path=req.video_path,
(API job path)\n    backend/main.py:1725 video_path=args.video_path, (CLI --video-path path)\n
Chain: emit_indexer_report (backend/reporting/_init_.py:62 video_path=video_path) ->
harvest.record_run (backend/reporting/harvest.py:227 video_path=video_path) ->
obsreport.RunRecorder (sets envelope.identity.video_path) -> rec.write() (sports-
obsreport/.../recorder.py:136 video_path=self.envelope.identity.video_path) ->
io.report_paths/_base (io.py:48-49). The worker is already correct:
backend/worker/_main_.py:298 passes video_path=local_video.\n\n(b) MINIMAL FIX (route through
the already-resolved local path ? zero new code, mirrors the worker)\n File:
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py\n Edit 1
(API path, line 953):\n    BEFORE: video_path=req.video_path,\n    AFTER:
video_path=local_video,\n Edit 2 (CLI path, line 1725):\n    BEFORE:
video_path=args.video_path,\n    AFTER: video_path=local_video,\n Why safe: `local_video` =
storage.acquire_local_input(...) (main.py:804 and :1658) returns a real LOCAL filesystem path ?
the bare stripped path for local:// (reports land next to the real file) and a scratch tempfile
under the system temp dir for gs://gcs://s3:// (reports land in scratch). For a bare/local input
it equals the original path -> byte-for-byte identical to today (no behaviour change for the
local workflow). DB provenance still records the SOURCE URI (db.add_video uses req.video_path /
args.video_path, unchanged). `local_video` is in scope at both finally blocks. Optional belt-
and-suspenders (not required): obsreport io.py:48-49 could skip the abspath-next-to-video branch
when re.match(r'^[A-Za-z][A-Za-z0-9+.\-]*://', video_path), falling through to the
reports/<video_id>/ branch ? but the host-side 2-line fix is sufficient and lower-risk; do not
edit the sibling repo unless desired.\n\n(c) .gitignore LINES to block recurrence\n File:
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.gitignore ? append a new
block (place after the \"Local generated artifacts\" block, ~line 50):\n    # Stray scheme-
prefixed report dirs from a URI accidentally used as a local path (issue #397).\n    # A belt-
and-suspenders guard so a regression can never ride into a commit; the real fix routes\n    #
reporting through the resolved local path (backend/main.py emit_indexer_report
video_path=local_video).\n    /gcs:/\n    /gs:/\n    /local:/\n    /s3:/\n    /gdrive:/\n
(Anchor each with a leading \"\" so only the REPO-ROOT scheme dirs are ignored, not
legitimately-named nested paths.)\n\n(d) EXISTING DIRS ? regenerable junk, SAFE TO DELETE
(verified)\n - Contents: 48 files total, all report artifacts only ? 16 *.report.json + 32
(*.report.md + *.summary.md). No source, no unique data.\n - Tracked by git? NO. `git ls-files
'gcs:*' 'gs:*' 'local:*'` returns nothing; `git status --porcelain` shows them as untracked (??
gcs:/, ?? gs:/, ?? local:/).\n - All regenerable: produced by running the offline test suite
(test_api_endpoints.py / test_reporting_harvest paths). Safe to remove:\n    rm -rf -- 'gcs:'
'gs:' 'local:' (run from the repo root)\n\nUNIT TEST TO ADD (assert no scheme-prefixed dir is
created at repo root)\n File: /home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_api_endpoints.py\n Add a regression test (mirrors the existing env fixture +
fake_fetch pattern). It posts a bucket URI AND a local:// URI through the real (un-mocked)
emit_indexer_report and asserts no `gcs:` `gs:` `local:` `s3:` dir appears at the repo root
(cwd). It FAILS on today's code and PASSES after the fix:\n\n    def
test_process_uri_input_writes_no_scheme_dir_at_repo_root(env, monkeypatch):\n        \"\"\"Issue
#397: a scheme URI must never become a literal 'gcs:/...'/'local:/...' dir at the\n        repo
root. The report writer must receive the RESOLVED local path, not the raw URI.\"\"\"\n
import os\n        client, db, tmp_path, mp = env\n        # A real on-disk file the local://
case can resolve+stat; bucket case is fetched to scratch.\n        real = _video(tmp_path,
\"ls.mp4\")\n        def fake_fetch(uri, dst=None, *, config=None, overwrite=False):\n
with open(dst, \"wb\") as fh: fh.write(b\"fetched\")\n        return dst\n
mp.setattr(m.storage, \"fetch\", fake_fetch)\n        mp.setattr(m.registry,
\"run_all_with_context\", lambda *a, **k: (_pass(), {}))\n
mp.setattr(m.segmenter_registry, \"get\", lambda name: _FakeSeg)\n        before =
set(os.listdir(\".\"))\n        for uri in (\"gs://bucket/videos/x.mp4\",
\"gcs://bucket/videos/x.mp4\",
\"local://\" + str(real).replace(\"\\\\\\\\\\\\\\\\\",
\"/\")):\n            r = client.post(\"/api/process\", json={\"video_path\": uri,
\"segmenter\": \"x\"})\n            assert r.status_code == 202\n
client.get(f\"/api/jobs/{r.json()['job_id']}\") # drain (inline executor)\n        new =
set(os.listdir(\".\")) - before\n        assert not any(d.rstrip(\"/\").endswith(\":\") or
\":/\" in d or\n            d.split(\"/\")[0] in {\"gcs:\", \"gs:\", \"local:\",
\"s3:\", \"gdrive:\"} for d in new), \"\n        f\"scheme-prefixed dir(s) leaked to repo
root: {sorted(new)}\"\n        for d in (\"gcs:\", \"gs:\", \"local:\", \"s3:\", \"gdrive:\"):

```

```

assert not os.path.isdir(d), f"{d} created at repo root"
Note: this test relies on obsreport being installed (it is, in this env). If CI may run without it, guard with
`pytest.importorskip("obsreport")` at the top of the test so it skips cleanly where the writer
is a no-op.",
    "files_touched": [
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.gitignore",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tests/test_api_endpoints.py"
    ],
    "new_or_updated_tests": [
        "tests/test_api_endpoints.py::test_process_uri_input_writes_no_scheme_dir_at_repo_root
(NEW) ? drives the API path with gs://, gcs://, and local:// URIs through the un-mocked
emit_indexer_report and asserts no gcs://gs://local://s3://gdrive: dir is created at the repo root
(cwd). Fails on current code, passes after main.py uses local_video."
    ],
    "coverage_note": "Investigated all emit_indexer_report callers:
backend/worker/__main__.py:298 (passes local_video ? already correct, unaffected),
backend/main.py:953 (API) and :1725 (CLI) (both pass the raw URI ? the two writers to fix),
backend/api/job_worker.py:78 (reconstructs a ProcessRequest and routes back through main.py, so
covered transitively). grep across backend/ for os.makedirs/open on uri-like strings found only
the worker scratch path (clean, routed through storage.put). Root cause reproduced empirically:
os.path.abspath('gcs://bucket/videos/x.mp4') -> '<repo_root>/gcs:/bucket/videos/x.mp4'.
Confirmed the fix yields clean dirs (next-to-file for local://, scratch tempdir for buckets) and
is identity for bare local paths. Verified the 48 stray files are untracked report artifacts
(json+md only), safe to delete. The fix is 2 one-line edits; .gitignore + the new test are the
belt-and-suspenders guard.",
    "risks_or_unknowns": "- Behaviour change for URI inputs is intentional and desirable:
report files now land next to the resolved local file / in scratch instead of in a bogus repo-
root scheme dir. For bare/local inputs there is ZERO change (local_video == the original path).
No DB/provenance change (source URI still stored via add_video).\n- For fetched bucket inputs,
reports land in the scratch tempdir, which is reclaimed by the OS ? same lifecycle as the worker
today; if a caller wants durable bucket-URI reports they should set config report.output_dir
(already supported via emit_indexer_report's output_dir path). This matches the worker's
existing semantics, so it is not a regression.\n- The new test and the existing bucket/local-
scheme API tests require obsreport installed to exercise the writer; in an env without it the
writer is a no-op (and never creates the dirs), so guard the new test with
pytest.importorskip('obsreport') for CI portability.\n- Optionally hardening obsreport
io.py:48-49 to detect a scheme:// and skip the next-to-video branch would defend other (e.g.
sibling-repo) callers, but is OUT of scope for this surgical host-side fix and touches a
different repo; left as a noted option only.\n- Deletion command must be run from the repo root
and the paths quoted (the trailing ':' is literal): rm -rf -- 'gcs:' 'gs:' 'local:'."
    }
},
{
    "key": "issue-399-links-convention",
    "data": {
        "summary": "Plan for issue #399 (launch-PR process hardening) in
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer, in four parts.\n\n(1)
BROKEN-LINK SCAN ? wrote + ran a stdlib-only scanner (saved at /tmp/claude-1000/-home-avidullu-
projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/scratchpad/check_md_links.py). It
parses every [text](target) in root *.md + docs/**/*.md (109 files), resolves relative paths
against each file's dir, and validates #anchors against GitHub-slugified headings. TOTAL = 52
broken internal links. Critically, 49 of the 52 are RELATIVE links inside docs/archives/** that
became historical-by-design when their projects were git-mv'd (DOC_STATUS.md §2 step 5:
"Relative links inside become historical"). Only 3 are in LIVE (non-archived) docs and MUST be
fixed in this PR:\n - docs/I18N_PLAN.md:4 -> #open-decisions-for-the-owner (anchor; heading is
"## 4. Open decisions (for the owner)" ? slug is "4-open-decisions-for-the-owner"; the
"4-" is missing)\n - docs/QUALITY_ITERATIONS.md:621 -> PER_VIDEO_OUTPUT_LAYOUT.md (doc was
archived+renamed 2026-06-21 to archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md)\n -
docs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md:10 ->
archives/research/AUDIO_RALLY_DETECTION_PLAN.md (file is under docs/archives/research/, but doc
was relocated to docs/data_pipeline/ so the relative base is now one level too deep ? needs
../)\nFull 52-entry list is in "exact_changes". The recommended PR scope is: fix the 3 live
breakages now (verified resolvable), and either (a) fix the 49 archived ../ links by adding one

```

more ../ level, or (b) accept them as historical per DOC_STATUS §2 and have the committed checker exclude docs/archives/** ? the plan recommends (b) + fix the 3.\n\n(2) CONVENTION codification ? there is NO .github/PULL_REQUEST_TEMPLATE.md or directory anywhere (only .github/workflows/ exists). Two coordinated edits: create .github/PULL_REQUEST_TEMPLATE.md with a launch/bit-flip checklist item, AND add a bullet to CLAUDE.md \"Workflow conventions\" (lines 64-70). Exact text + locations in \"exact_changes\".\n\n(3) CI LINK-CHECKER ? propose committing the scanner as tools/check_md_links.py (stdlib only, mirrors the existing tools/leak_scan.py pattern, run via `python -m tools.check\_md\_links`) and adding a \"Markdown link check\" job to .github/workflows/ci.yml. Script sketch + workflow step in \"exact_changes\". Recommend the committed checker exclude docs/archives/** (historical-by-design) so it fails only on live-doc breakages.\n\n(4) Rule?code cross-link ? in docs/RALLY_DETECTION_FIXES_PLAN.md §8 Q2, the \"(promote iff tolF1 does not regress)\" rule should link to backend/eval/promotion.py (which exists; defines served_regresses + promote_if_served_safe). Exact edit in \"exact_changes\".

\"exact_changes\": \"=====\nPART 1 ?

BROKEN-LINK SCAN: full list (52 total)\n=====\nScanner: stdlib-only, parses [text](target) in root *.md + docs/**/*.md, resolves paths\nrelative to each md file, slugifies ATX headings GitHub-style to validate #anchors.\nSaved at: /tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/scratchpad/check_md_links.py\n(Note: gs:/, gcs:/, local:/ pytest-artifact dirs are intentionally NOT scanned ? scope is root + docs/.)\n\nTOTAL broken internal links: 52 (scanned 109 markdown files)\n-> 3 in LIVE docs (MUST FIX), 49 inside docs/archives/** (historical-by-design per DOC_STATUS §2 step 5)\n\n-- A. LIVE docs ? MUST FIX in this PR (3) ---\ndocs/I18N_PLAN.md:4 -> #open-decisions-for-the-owner [missing anchor (same file)]\ndocs/QUALITY_ITERATIONS.md:621 -> PER_VIDEO_OUTPUT_LAYOUT.md [missing path]\ndocs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md:10 -> archives/research/AUDIO_RALLY_DETECTION_PLAN.md [missing path]\n\nFixes (all verified to resolve):\n ? docs/I18N_PLAN.md:4\n BEFORE: ...see [Open decisions](#open-decisions-for-the-owner).\n AFTER: ...see [Open decisions](#4-open-decisions-for-the-owner).\n (heading \"### 4. Open decisions (for the owner)\" slugifies to 4-open-decisions-for-the-owner)\n ? docs/QUALITY_ITERATIONS.md:621\n BEFORE: (PER_VIDEO_OUTPUT_LAYOUT.md)\n AFTER: ([PER_VIDEO_OUTPUT_LAYOUT.md](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md))\n ? docs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md:10\n BEFORE: ...[AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md)\n AFTER: ...[AUDIO_RALLY_DETECTION_PLAN.md](../archives/research/AUDIO_RALLY_DETECTION_PLAN.md)\n\n-- B. docs/archives/** ? historical-by-design (49) ---\n\n(DOC_STATUS.md §2 step 5: \"Relative links inside become historical (note this in the doc's header).\" The DECOUPLED_COMPUTE set moved to past_projects/ so its ../ links are now one level too shallow.)\ndocs/archives/COMMERCIAL_READINESS_REVIEW.md:12 -> BACKLOG.md\ndocs/archives/field-pmf/PMF_MARKET_RESEARCH.md:6 -> COMMERCIALIZATION.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:16 -> ../PLATFORM_ARCHITECTURE.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:30 -> ../HOSTING_PLAN.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:38 -> ../NEXT_STEPS.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:42 -> ../BACKLOG.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:46 -> ../ROADMAP.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:57 -> ../OWNED_MODEL_IMPLEMENTATION_PLAN.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:60 -> ../training/README.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:77 -> ../GOLDEN_DATA_SHARING.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:86 -> ../backend/pipeline/detectors/base.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:88 -> ../backend/main.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:89 -> ../backend/pipeline/segmenters/trajectory_hybrid.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md:94 -> ../sports-data-collector\ndocs/archives/past_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md:15 -> ../backend/pipeline/detectors/base.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md:16 -> ../training/gen0/harness.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md:28 -> ../backend/pipeline/segmenters/trajectory_hybrid.py\ndocs/archives/past

```

_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md:30 ->
../../backend/main.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md:17
-> ../PLATFORM_ARCHITECTURE.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-
PLAN.md:30 ->
../PLATFORM_ARCHITECTURE.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-
PLAN.md:42 ->
../../backend/eval/regression.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-
TEST-PLAN.md:83 ->
../../backend/eval/golden_fixtures.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-
AND-TEST-PLAN.md:107 -> ../../backend/pipeline/detectors/base.py\ndocs/archives/past_projects/DE
COUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md:116 ->
../DESKTOP_APP_PLAN.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-
PLAN.md:141 -> ../../backend/pipeline/segmenters/trajectory_hybrid.py\ndocs/archives/past_projec
ts/DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md:7 ->
../../backend/infrastructure/database.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/05-COST-
ATTRIBUTION.md:68 ->
../PLATFORM_ARCHITECTURE.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/06-RISKS-AND-OPEN-
QUESTIONS.md:16 -> ../../backend/pipeline/detectors/base.py\ndocs/archives/past_projects/DECOUPL
ED_COMPUTE/07-COMPUTE-ABSTRACTION.md:14 ->
../../backend/config/models.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/07-COMPUTE-
ABSTRACTION.md:15 -> ../../backend/pipeline/detectors/person_detector.py\ndocs/archives/past_pro
jects/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md:18 -> ../../backend/pipeline/detectors/base.py
\ndocs/archives/past_projects/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md:20 ->
../../backend/utils/run_telemetry.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:41
-> ../PLATFORM_ARCHITECTURE.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:43 ->
../HOSTING_PLAN.md\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:60 -> ../../backend/
eval/golden_fixtures.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:113 -> ../../ba
ckend/infrastructure/database.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:113 ->
../../backend/api/job_worker.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:115 ->
../../training/gen0/harness.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:116 -> .
../../backend/eval/golden_fixtures.py\ndocs/archives/past_projects/DECOUPLED_COMPUTE/README.md:11
7 -> ../../backend/pipeline/segmenters/trajectory_hybrid.py\ndocs/archives/quality-
iterations/2026-06-18-lovo-resweep-n13/README.md:8 ->
../../../../GOLDEN_VIDEOS.md\ndocs/archives/quality-iterations/2026-06-18-lovo-
resweep-n13/README.md:74 -> ../../../../GOLDEN_VIDEOS.md\ndocs/archives/roora-
vendor/GOLDEN_SET_VENDORING.md:8 -> EVALUATION.md\ndocs/archives/roora-
vendor/GOLDEN_SET_VENDORING.md:8 -> archives/decisions/DECISIONS.md\ndocs/archives/roora-
vendor/GOLDEN_SET_VENDORING.md:8 -> VIDEO_RECORDING_GUIDELINES.md\ndocs/archives/roora-
vendor/GOLDEN_SET_VENDORING.md:8 -> EXPERIMENT_HARNESS.md\ndocs/archives/roora-
vendor/GOLDEN_SET_VENDORING.md:10 -> GOLDEN_SET_IMPLEMENTATION.md\ndocs/archives/roora-
vendor/ROORA_LABELING_GUIDELINES.md:11 -> MULTI_SIGNAL_FUSION_PLAN.md\ndocs/archives/roora-
vendor/ROORA_LABELING_GUIDELINES.md:136 ->
MULTI_SIGNAL_FUSION_PLAN.md\n\n=====
? CONVENTION: codify the archive-on-launch
rule\n\n=====
\nFinding: NO PR template exists
(searched whole repo; .github/ contains only workflows/).\nSo codify in TWO places ? a new PR
template (process gate) + CLAUDE.md (agent memory).\n\n(2a) NEW FILE:
.github/PULL_REQUEST_TEMPLATE.md ? add this checklist
block:\n-----\n\n## Launch /
bit-flip PRs ? archival gate\n\nIf this PR **launches** a tracked project or **flips a default**
(a \"bit-flip\" promotion, ne.g. a flag ? default-ON), you MUST also run the
`docs/DOC_STATUS.md` §2 archival\ndue-diligence checklist for the affected project doc(s):\n- [
] Not a launch/bit-flip PR ? gate N/A, **or** all boxes below are checked.\n- [ ] **Verified**
the project doc's claims against the code (`[verified]` load-bearing facts).\n- [ ] **Flipped
Status ? DELIVERED/DONE/REJECTED** + a completion note (what shipped, where, deferred).\n- [ ]
**Updated all inbound refs** (`docs/README.md`, `CLAUDE.md`, `docs/NEXT_STEPS.md`, cross-doc
links).\n- [ ] **Kept the lineage** (ADR / supersedes links) and **`git mv` the doc ?
`docs/archives/`** (terminal-state only).\n- [ ] **Recorded the move** ? flipped its row in
`docs/DOC_STATUS.md` §3 and noted it in memory `current-
state`.\n-----\n\n(2b) EDIT
CLAUDE.md \"Workflow conventions\" (the section currently at lines 64-70).\nAdd a new bullet
right after the existing PROJECT_DOC_TEMPLATE bullet (currently line
68):\n-----\n\n- **A launch /
bit-flip PR MUST archive the project doc(s) it completes.** When a PR launches a\n tracked
project or flips a default (promotion / bit-flip), run the `docs/DOC_STATUS.md` §2 archival\n

```

due-diligence checklist in the SAME PR: verify-vs-code ? flip Status + completion note ?
update inbound refs ? keep lineage ? `git mv` to `docs/archives/` ? record the move in §3.
Enforced by the `github/PULL\_REQUEST\_TEMPLATE.md` archival gate.

-----\n(Rationale: CLAUDE.md \$\"Workflow conventions\" + \$\"Read these first\" already point to DOC_STATUS §2 as the archive process; this bullet makes the launch/bit-flip TRIGGER explicit and ties it to the PR gate.)\n\n===== \nPART 3 ? CI LINK-CHECKER (lightweight, dependency-free)\n===== \nCommit the scanner as tools/check_md_links.py (mirrors the existing `python -m tools.leak\_scan` pattern; stdlib only ? re, pathlib, sys). Recommended: EXCLUDE docs/archives/** (historical-by-design per DOC_STATUS §2 step 5) so CI fails only on live-doc breakage. The working script is at the scratchpad path above; add an `--exclude docs/archives` flag (or hardcode the prefix skip). Script sketch:\n\n# tools/check_md_links.py (stdlib only)\nLINK_RE = re.compile(r\"(?![\\!\\])\\[([\\^\\]]*)\\]\\[([\\^\\])+(\\^\\)]\\)\")\nATX = re.compile(r\"^\\s{0,3}(#{1,6})\\s+(.*)\\s*#\\s*\$\")\ndef slugify(h): # strip code/format markers, lower, drop non-[word/space/-], spaces->'-' , dedupe -1,-2\n def collect_anchors(md): # ATX headings (skipping `` fences) + -> set of slugs\n def main():\n files = sorted(repo.glob(\"**/*.md\")) + sorted((repo/\"docs\").rglob(\"**/*.md\"))\n files = [f for f in files if \"docs/archives/\" not in f.as_posix()] # historical-by-design\n for md in files:\n for lineno, target in iter_links(md): # skip fenced code + http(s)/mailto + absolute-URL schemes\n path, _, anchor = target.partition(\"#\")\n if path == \"\": check anchor in collect_anchors(md)\n else: resolved = (md.parent/path).resolve(); check exists; if .md and anchor: check slug\n print each\n \"file:line -> target [reason]\"; print TOTAL; return 1 if broken else 0\n if __name__ == \"__main__\": raise SystemExit(main())\n\nWorkflow step ? add this job to .github/workflows/ci.yml (sibling of the existing `leak-scan` job; no pip install needed since it's stdlib-only):\n\n# Internal-doc link integrity (tools/check_md_links.py): fails on a broken relative path or a\n# dead #anchor in any LIVE markdown (root + docs/, excluding docs/archives/** which is\n# historical-by-design per DOC_STATUS §2). Stdlib-only ? no deps to install.\nlink-check:\n name: Markdown link check (internal)\n runs-on: ubuntu-latest\n timeout-minutes: 10\n steps:\n - uses: actions/checkout@v6\n - uses: actions/setup-python@v6\n with:\n python-version: \"3.13\"\n - name: Check internal markdown links\n run: python -m tools.check_md_links\n\n(Optional hardening: also commit tests/test_check_md_links.py with tmp_path fixtures asserting a missing path, a dead anchor, and a valid link ? so the checker itself is regression-guarded.)\n\n===== \nPART 4 ? rule?code cross-link in RALLY_DETECTION_FIXES_PLAN.md\n§8\n===== \nbackend/eval/promotion.py EXISTS (defines `served\_regresses` and `promote\_if\_served\_safe`, `\_\_all\_\_` on line 35). In §8 Q2 the rule phrase appears twice; link it where the rule is stated in §8 (line 110). Edit docs/RALLY_DETECTION_FIXES_PLAN.md line 110: BEFORE: `promote\_if\_served\_safe(structural=True) ? promote`. Per the rule (promote iff tolF1 does not) AFTER: `promote\_if\_served\_safe(structural=True) ? promote`. Per the rule ([promote iff tolF1 does not regress](../backend/eval/promotion.py) ? `served\_regresses` / `promote\_if\_served\_safe`)\n\n(Note: the phrase wraps onto line 111 \"...regress) ? **PROMOTED**...\"; fold the linked text onto one markdown link spanning 110-111. The path is ../backend/eval/promotion.py ? verified to exist relative to docs/.)\n\nOptionally also link the prescriptive statement on line 99 (\"...a one-line PR) **only if tolF1 does not regress**.\") the same way.,\n\n\"files_touched\": [\n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/I18N_PLAN.md\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/QUALITY_ITERATIONS.md\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/RALLY_DETECTION_FIXES_PLAN.md\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/CLAUDE.md\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.github/PULL_REQUEST_TEMPLATE.md\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.github/workflows/ci.yml\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tools/check_md_links.py\", \n \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/tools/README.md\n]

(optional: document the new tool)"

```
],
  "new_or_updated_tests": [
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/tests/test_check_md_links.py (NEW, optional but recommended): tmp_path fixtures
asserting (a) a missing relative path is flagged, (b) a dead #anchor is flagged, (c) a valid
path+anchor passes, (d) http/mailto/absolute and fenced-code links are skipped, (e)
docs/archives/** is excluded. Mirrors the offline/stdlib style of the existing suite."
  ],
  "risks_or_unknowns": "- This is a PLAN; no repo files were modified. The scanner was
written/run only in the scratchpad.\n- Issue #399 text was not read directly (no issue tracker
access in this no-git-remote checkout); scope is inferred from the prompt + DOC_STATUS.md $2 and
CLAUDE.md conventions. Confirm the issue means \"fix all 52\" vs \"fix the 3 live ones\" ? I
recommend the latter for must-fix, with the 49 archived ../ links treated as historical-by-
design (DOC_STATUS $2 step 5) and excluded by the committed CI checker. If the owner wants the
archived links fixed too, the DECOUPLED_COMPUTE set (moved one level deeper into past_projects/)
needs one extra ../ per link, plus a few archived files reference docs that themselves moved
(e.g. roora-vendor -> EVALUATION.md/EXPERIMENT_HARNESS.md need ../ to docs/).\n- The slugify()
in the scanner is a good-enough subset of GitHub's algorithm (handles ATX headings, fenced-code
skipping, duplicate-slug -1/-2, inline code/emphasis stripping, <a name=/id=>). It does NOT
special-case emoji or every Unicode edge; the repo uses emoji in some headings (e.g. DOC_STATUS
\"? LIVING\") ? but no live link targets those anchors, so no false positive was produced. If a
future link targets an emoji heading, refine slugify to drop emoji to match GitHub.\n- Scanner
deliberately scopes to root *.md + docs/**; it excludes the gs:/ gcs:/ local:/ pytest-artifact
dirs and backend/deploy/scripts READMEs. Confirm whether the CI checker should also cover those
nested READMEs (e.g. scripts/README.md, deploy/**/README.md) ? currently out of scope per the
issue wording (docs/ + root).\n- The $8 cross-link wraps across two source lines (110-111); the
editor must fold the markdown link text across the wrap or the link will only cover part of the
phrase.\n- PR-template placement: GitHub honors .github/PULL_REQUEST_TEMPLATE.md (single default
template). If multiple templates are later wanted, move to .github/PULL_REQUEST_TEMPLATE/
directory form.",
  "coverage_note": "Scanner ran over all 109 markdown files in root + docs/ (the
gs:/gcs:/local: pytest-artifact dirs and backend/deploy/scripts/eval_baselines READMEs are out
of the issue's stated scope and were excluded). All 3 live-doc fixes were verified to resolve
(target files exist; computed slug 4-open-decisions-for-the-owner matches the heading).
backend/eval/promotion.py confirmed present with served_regresses + promote_if_served_safe.
Confirmed there is NO existing PR template and NO existing link-check CI job, so both Part 2 and
Part 3 are net-new. DOC_STATUS.md $2 (the archival checklist) and CLAUDE.md \"Workflow
conventions\" were read in full to ground Part 2's exact wording."
}
},
{
  "key": "doc-archival-plan",
  "data": {
    "archive_ready_docs": [],
    "not_ready_docs": [
      "docs/RALLY_DETECTION_FIXES_PLAN.md ? Status: DONE (all 3 findings A/B/C resolved; P2b
promoted #396). DESPITE being terminal in its own $7, it is NOT archive-ready: its own header
(lines 8 & 30) explicitly DEFERS archival 'until the wider R-series quality cluster is terminal,
per docs/DOC_STATUS.md.' It is the FIXES sub-track of the gated R-cluster (lineage: extends
RALLY_DETECTION_QUALITY_REPORT.md, references RALLY_DETECTION_CODE_REVIEW.md). GATE = the
cluster gate below.",
      "docs/RALLY_DETECTION_CODE_REVIEW.md ? the original 2026-06-19 review snapshot for
FIXES_PLAN's findings A/B/C, carrying a 2026-06-29 resolution note. Terminal in content but tied
to FIXES_PLAN; archive WITH FIXES_PLAN when the cluster wraps. Same cluster gate. (Note: not
currently tracked in DOC_STATUS $3 at all ? see findings.)",
      "docs/RALLY_QUALITY_RESEARCH.md ? Foundation/tracked-project, ACTIVE. GATE: $7.4 DoD not
met ? $7.2 Tier-1/2 items still open (W1.5 ?, W2 ? multi-feature scorer 'next (GPU)', M1 ?, M2
?) and $7.1's single quality number / gap-closure % is still 'TODO' (un-attributed). Complete
only when all committed tiers land AND the single number is recorded.",
      "docs/R2_EVAL_METRIC_DESIGN.md ? IMPLEMENTED (augmenting). GATE: tolF1 is NOT yet
promoted-or-rejected as THE gate. $6/$7 still hold tIoU as the gate; R2 augments until a
dedicated ablation experiment on the golden corpus validates it. NEXT_STEPS.md:21 still lists
'promote-or-reject R2 tolF1 as the gate (the ablation)' as a pending owner-side GPU item. P2b
did NOT do this ablation ? it promoted a different flag.",
    ]
  }
}
```

```

"docs/QUALITY_ITERATIONS.md ? LIVING empirical log / paper-backbone ablation log
([[paper-coauthor-aim]]). DOC_STATUS §3c says 'Keep'. NEVER archive as a whole; only terminal-
state entries graduate to archives/quality-iterations/. Not a cluster-archive target.",
"docs/R1_MILESTONE_LAUNCH_REPORT.md ? LAUNCHED (#204), terminal record. GATE: kept with
the R-series for narrative; archive only when the whole track wraps (same cluster gate).",
"docs/RALLY_DETECTION_GAP_CLOSURE.md ? DRAFT, implementation owner-gated (§7 G1?G3
levers not started). Not terminal; part of the R/quality cluster.",
"docs/REAL_FOOTAGE_VALIDATION_REPORT.md ? 'validation complete, roadmap proposed' ?
motivates R2; archive WITH the R-cluster (same gate), not standalone.",
"docs/RALLY_DETECTION_QUALITY_REPORT.md ? report/quality snapshot (checkpoint
2026-06-17). Part of the R-cluster; archive as a cluster once the gate clears. Not terminal
alone.",
"docs/PROMINENT_COURT_DETECTION.md ? Design+experiment, default-OFF, owner-gated flip
(G2 over-fire). Part of the rally/quality cluster; not terminal.",
"docs/HOW_RALLY_DETECTION_WORKS.md ? LIVING reference (2-layer mental model); per
DOC_STATUS §3b 'Keep current; not an archive target.'"
],
"per_doc_inbound_refs": "No doc is archive-ready, so there are no inbound refs to rewire
for an archive move yet. For completeness, the inbound references to the two terminal-but-gated
FIXES docs (to be rewired only WHEN the cluster is archived)
are:\n\nRALLY_DETECTION_FIXES_PLAN.md (2 inbound):\n- docs/RALLY_DETECTION_CODE_REVIEW.md:9 ?
'See `docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status...' (intra-cluster; both move
together so the relative link stays valid)\n- docs/NEXT_STEPS.md:25 ? item 9b references
'finding B / P2 in `RALLY_DETECTION_FIXES_PLAN.md`'. NOTE: this line is now STALE ? it still
says PR #393 'default-OFF, to ship: golden-set A/B... flip... only if tolF1 doesn't regress',
but #396 already DID the A/B and promoted it default-ON. Update/remove item 9b (and re-check
item 8 line 21) regardless of archival.\n- (NOT in docs/README.md: FIXES_PLAN's own header
claims it is 'indexed in docs/README.md' but grep of README.md finds NO entry ? that self-claim
is itself stale.)\n\nRALLY_DETECTION_CODE_REVIEW.md (1 inbound):\n-
docs/RALLY_DETECTION_FIXES_PLAN.md:129 ? §10 References:
'docs/RALLY_DETECTION_CODE_REVIEW.md' (intra-cluster).\n- (No README.md / CLAUDE.md /
NEXT_STEPS.md inbound.)\n\nR-cluster docs that DO have README/index inbounds to rewire when the
cluster eventually archives: RALLY_QUALITY_RESEARCH.md (README.md:1),
RALLY_DETECTION_GAP_CLOSURE.md (README.md:2), RALLY_DETECTION_QUALITY_REPORT.md (README.md:4),
QUALITY_ITERATIONS.md (README.md:7), EXPERIMENT_HARNESS.md (README.md:6), plus DOC_STATUS.md
§3a/§3c/§4 rows for each ? but NONE move now.",
"steps": "No archive-ready docs ? run NO §2 steps and do NO git mv right now. When the
cluster gate clears, run §2 (steps 1-6) once for the WHOLE R-cluster as a unit (DOC_STATUS §4
finding #3 mandates 'archive as a cluster'):\n1. VERIFY claims vs code, tag [verified].
FIXES_PLAN+CODE_REVIEW are already verified/DONE; re-confirm the others
(QUALITY_REPORT/RESEARCH/R2/GAP_CLOSURE/REAL_FOOTAGE/R1) carry final measured numbers.\n2. Flip
Status ? DELIVERED/DONE/REJECTED + completion note on each (FIXES_PLAN+CODE_REVIEW already DONE;
the rest flip only when their §7/§8 DoD is met).\n3. Update ALL inbound refs (the
README.md:1/2/4/7, NEXT_STEPS.md items 8/9/9b, DOC_STATUS §3a/§3c/§4 rows) ? new archive path +
status.\n4. Keep lineage (FIXES_PLAN?extends?QUALITY_REPORT, FIXES_PLAN?CODE_REVIEW;
supersedes/superseded-by links).\n5. git mv the cluster ? docs/archives/past_projects/ (a single
rally-quality subdir).\n6. Record the move in DOC_STATUS §3e + memory current-
state.\n\nINDEPENDENT of archival (do now): (a) DOC_STATUS §3 does NOT yet track
RALLY_DETECTION_FIXES_PLAN.md or RALLY_DETECTION_CODE_REVIEW.md ? add rows (FIXES_PLAN =
DONE/gated-to-R-cluster; CODE_REVIEW = terminal snapshot/gated). (b) Fix the stale
NEXT_STEPS.md:25 item 9b (P2b/#396 already promoted the flag default-ON). (c) Fix FIXES_PLAN's
own header self-claim that it's 'indexed in docs/README.md' (it is not).",
"doc_status_protocol": "Per DOC_STATUS.md: §1 lifecycle legend (DRAFT?IN
PROGRESS?DONE/DELIVERED?ARCHIVED, plus LIVING/PROPOSAL/REPORT/CONVENTION). §2 = the 6-step
archival due-diligence checklist (verify?flip status?update inbound refs?keep lineage?git mv to
docs/archives/past_projects/?record the move) ? 'do NOT just git mv'. §4 finding #3 is the
binding gate for this whole cluster: 'R-series/quality cluster is ACTIVE ? Do NOT archive yet ?
archive as a cluster once R2 is promoted-or-rejected as the gate + §7.1 single number is
attributed (owner confirmed leave-as-is 2026-06-21).' P2b (#396) promoted the operational
`inactivity_temporal_density` flag (a do-no-harm correctness fix, NO measured lift ? ?tolF1
+0.010, 95% CI [?0.014,+0.034] spans 0, p=0.18); it is NOT the R2 tolF1 metric-gate ablation and
NOT the §7.1 number attribution. Therefore P2b does NOT close the cluster gate, and FIXES_PLAN ?
though DONE ? is gated to the cluster by its own header."
}
}

```

]

Output a tight Markdown plan: for EACH of the 5 PRs (#397 stray-dirs, #398 served-gate, #399 process-hardening, reportlab-uncap, doc-archival) give: the exact files to change, the test(s) to add/keep, coverage impact, risk level, and the recommended MERGE ORDER (lowest-risk first; put doc-archival last since it touches many inbound refs). Flag anything that should NOT be done (e.g. docs that are not archive-ready). Be concrete and command-accurate.

2. assistant (2026-06-29T17:05:50.493Z)

I'll start by verifying the key claims against the actual repo state, then produce the merged plan. Let me check the critical facts in parallel.

3. user (2026-06-29T17:05:56.335Z)

```
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer
---REPO---
master
---STRAYDIRS---
gcs:
gs:
local:
---RUNALL---
run_all_tests.py
---CI---
.github/workflows/ci.yml
---PRTEMPLATE---
no PR template (expected)
```

4. user (2026-06-29T17:05:57.690Z)

```
---reportlab refs---
---reportlab in code---
backend/eval/ablation.py
backend/eval/ablation_pdf.py
```

5. assistant (2026-06-29T17:06:00.553Z)

The `reportlab-uncap` item isn't in the provided JSON results ? it's a 5th PR mentioned only in the output instructions. Let me find where reportlab is pinned/capped.

6. user (2026-06-29T17:06:04.610Z)

```
===all requirements files===
requirements.txt
./requirements.txt
./training/requirements-train.txt
===grep reportlab everywhere===
backend/eval/ablation_pdf.py:1:"""reportlab PDF renderer for the ablation report (split out of
`backend.eval.ablation`).
backend/eval/ablation_pdf.py:4:booktabs-style PDF. `reportlab` is imported lazily inside
`export_pdf` so importing this
backend/eval/ablation_pdf.py:8:Kept as its own module so the ~116-line reportlab builder doesn't
bloat the pure-driver
backend/eval/ablation_pdf.py:28: """Render the `AblationReport` to a clean reportlab PDF: a
table of each signal's marginal
backend/eval/ablation_pdf.py:32: Returns the written path. `reportlab` is an offline, pure-
Python dependency (no GPU)."""
backend/eval/ablation.py:527:# The ~116-line reportlab PDF builder lives in a sibling module to
keep this file a pure driver;
tests/test_ablation.py:400:     pytest.importorskip("reportlab")
tests/test_ablation.py:629:     pytest.importorskip("reportlab")
```

7. assistant (2026-06-29T17:06:06.704Z)

The reportlab pin isn't in requirements.txt directly. Let me look more carefully ? it may be a capped/pinned version somewhere, or a transitive constraint.

8. user (2026-06-29T17:06:09.947Z)

```
===requirements.txt (reportlab + nearby)===
```

9. user (2026-06-29T17:06:14.359Z)

```
1      fastapi>=0.111.0
2      uvicorn>=0.30.1
3      opencv-python>=4.9.0.80
4      numpy>=1.26.4
5      pydantic>=2.7.1
6      jinja2>=3.1.4
7      python-dotenv>=1.0.0
8      google-genai>=2.4.0
9      # torch/torchvision are listed here but deliberately NOT in pyproject.toml
10     # [project.dependencies]. Their CPU/CUDA wheels need an out-of-band index
11     # (--index-url / --extra-index-url) that PEP 621 cannot express. Install
12     # them separately matching your hardware:
13     # NVIDIA GPU: pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu124
14     # CPU only: pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
15     # CI installs CPU torch on its own line in .github/workflows/ci.yml.
16     torch>=2.12.0
17     torchvision>=0.27.0
18     supervision>=0.21.0
19     # D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone
20     # `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see
21     # https://github.com/roboflow/trackers/issues/474), so install from git until fixed:
22     # pip install "trackers @ git+https://github.com/roboflow/trackers.git@2.5.0"
23     # (Apache-2.0; depends on supervision >=0.26.1, so it composes cleanly.)
24     # NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
25     # PIN-DUPLICATION (mirrors the torch carve-out): this exact ref is ALSO hard-coded in
26     # .github/workflows/ci.yml + nightly.yml (full-suite lanes) and
deploy/digitalocean/bootstrap.sh
27     # + deploy/cloudrun/Dockerfile ? edit ALL of them in lockstep on a bump, or the
lanes/boxes
28     # silently resolve different versions.
29     trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
30     scipy>=1.10
31
32     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
33     # no-op if absent). Shared, permissively-licensed report toolkit. Pin a tag once
34     # the private repo is published, e.g.:
35     # obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
36     # For local development install it editable from a sibling checkout:
37     # pip install -e ../sports-obsreport --no-build-isolation
38
```

10. assistant (2026-06-29T17:06:18.281Z)

The reportlab cap isn't in requirements.txt. Let me check pyproject.toml, the CI workflows, and the obsreport sibling repo (the soft dep that pulls reportlab), plus check `ablation\_pdf.py` for a version constraint.

11. user (2026-06-29T17:06:24.810Z)

```
===pyproject reportlab/deps===
```

```
8:# torch/torchvision are deliberately NOT listed in [project.dependencies]:
10:# which PEP 621 cannot express. Install them separately ? see [project.optional-dependencies]
27:dependencies = [
```

```
47:[project.optional-dependencies]
===grep reportlab in workflows + deploy===
===grep reportlab in sibling obsreport===
```

12. user (2026-06-29T17:06:27.498Z)

```
27     dependencies = [
28         "fastapi>=0.111.0",
29         "uvicorn>=0.30.1",
30         "opencv-python>=4.9.0.80",
31         "numpy>=1.26.4",
32         "pydantic>=2.7.1",
33         "jinja2>=3.1.4",
34         "python-dotenv>=1.0.0",
35         "google-generativeai>=2.4.0",
36         "supervision>=0.21.0",
37         # Used directly on the serving path (backend/features/rally_seq.py: uniform_filter1d
38         # maximum_filter1d); previously present only TRANSITIVELY via supervision, so a
39         # bump or a pruned-transitive freeze could silently break inference. Declare it.
40         # (BSD-3.)
41         "scipy>=1.10",
42         # D13/P12: ByteTrack moved to the standalone `trackers` package (Apache-2.0). Its
43         # PyPI wheel is broken (metadata-only), so it MUST be installed from git ? see
44         # requirements.txt. PEP 621 cannot express a git URL, so it's NOT listed here
45         # (same pattern as torch/torchvision). CI installs it on its own line.
46     ]
47     [project.optional-dependencies]
48     # Test + lint/type tooling ? matches the CI lanes (.github/workflows/ci.yml).
49     dev = [
50         "pytest",
51         "pytest-cov",
52         "httpx",          # fastapi TestClient transport
53         "ruff",
54         "mypy",
55         "PyJWT[crypto]>=2.8", # M9 edge-auth: Cf-Access JWT verify (RS256) ? tests + CI
56     ]
57     # M9 edge-auth (Cloudflare Access). DEFAULT-OFF: only needed when
58     security.edge_auth_enabled=True.
59     # `jwt` is lazy-imported, so the base runtime works without this; install it for
60     hosted/multi-tenant.
61     auth = ["PyJWT[crypto]>=2.8"]
62     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
63     # no-op if absent). Pinned to its git tag per BACKLOG. The private repo is
64     # fetched over SSH; if you lack access, simply omit this group ? reporting
65     # tests skip via pytest.importorskip. For local dev against a sibling checkout:
66     # pip install -e ../sports-obsreport --no-build-isolation
```

13. assistant (2026-06-29T17:06:32.831Z)

reportlab is not pinned/capped anywhere in this repo's declared deps. It's pulled transitively via the soft `obsreport` dependency. Let me check the obsreport sibling's pyproject for the reportlab cap, and check the open issue/backlog text for "reportlab" to understand what "reportlab-uncap" means.

14. user (2026-06-29T17:06:37.567Z)

```
===obsreport pyproject/setup===
sports-obsreport/pyproject.toml
---obsreport reportlab pin---
```

```
===grep 'reportlab' in indexer docs/backlog===
badminton-highlight-indexer/docs/ABLATION_LAYER.md:61:reportlab report (table of marginal
contribution + CI + sign test + a verdict column).
===grep uncap/cap reportlab anywhere===
badminton-highlight-indexer/docs/ABLATION_LAYER.md:61:reportlab report (table of marginal
contribution + CI + sign test + a verdict column).
```

15. assistant (2026-06-29T17:06:41.627Z)

The `reportlab-uncap` work item has no trace in this repo's pinned deps, the obsreport sibling, docs, or the four investigation JSONs. It was named only in my output instructions with no investigation result attached. Let me do a final broad search to be certain there's no version cap anywhere (e.g. a constraints file or a pinned `reportlab<X` somewhere I haven't looked), and check the ablation_pdf import guard.

16. user (2026-06-29T17:06:46.286Z)

```
===any version-capped reportlab string repo-wide===
===constraints files===
===reportlab in obsreport pyproject full===
6:name = "obsreport"
8:description = "Reusable per-run observability report toolkit: a small config-language for
emitting technical + customer-friendly reports across data/video pipelines."
13:dependencies = [
19:[project.optional-dependencies]
===ablation_pdf import line===
1:"""reportlab PDF renderer for the ablation report (split out of ``backend.eval.ablation``).
4:booktabs-style PDF. ``reportlab`` is imported lazily inside ``export_pdf`` so importing this
8:Kept as its own module so the ~116-line reportlab builder doesn't bloat the pure-driver
28:    """Render the ``AblationReport`` to a clean reportlab PDF: a table of each signal's
marginal
32:    Returns the written path. ``reportlab`` is an offline, pure-Python dependency (no
GPU)."""
33:    from reportlab.lib import colors
34:    from reportlab.lib.enums import TA_CENTER, TA_LEFT
35:    from reportlab.lib.pagesizes import A4
36:    from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
37:    from reportlab.lib.units import cm
```

17. assistant (2026-06-29T17:06:50.416Z)

Confirmed: there is no `reportlab` version cap anywhere in this repo (no pin in `requirements.txt`, `pyproject.toml`, workflows, deploy, constraints files, or the `obsreport` sibling). The `reportlab-uncap` PR has no investigation result in the provided JSON and no actionable target I can verify in the tree. I'll flag it explicitly rather than fabricate file changes.

Now let me verify a few load-bearing facts from the three investigations that have JSON data, so the plan's commands are accurate.

18. user (2026-06-29T17:06:55.197Z)

```
===398: served_gate_a key lines===
from backend.eval import segmentation_metrics as _segm
from backend.eval.calibrate_wasb import load_golden_gts
from backend.eval.metrics import Interval, Score, score
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter

def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:
    """The ``indexing`` config dict the served FusionSegmenter seems read ? mirrors
    config.json's ``indexing.fusion`` block, with Gate A at ``min_crossings`` and the
    person cue OFF (so no GPU / torch is touched)."""
    return {
```

```

    "fusion": {
        "substrate": "wasb",
        "min_crossings": min_crossings,
        "min_players": 0,
        "cue_flags": {},
    },
    "wasb": {"velocity_threshold": PROD_VELOCITY_THRESH},
    "dead_time_merge_gap": PROD_MERGE_GAP,
    "min_action_window_duration": PROD_MIN_WINDOW,
}
idx_cfg = _served_idx_cfg(min_crossings)
# Construct the real served segmenter (db=None, config carries the indexing block). We only
# call the pure decision seams ? no process_video, no DB, no provider ? exactly as the
# registry constructs it for `name`/`description`.
seg = FusionSegmenter(None, {"indexing": idx_cfg, "sport": "badminton"})
# The served per-window stats (incl. net_crossings) via the production enrichment hook.
idx_cfg_typed: Any = idx_cfg # type: ignore[annotation-type-mismatch]
window_stats = [
    seg._enrich_candidate_stats(
        cw, points, fps, frame_width, video_path="", idx_cfg=idx_cfg_typed
    )
    for cw in windows
]
keep = seg._select_for_handoff(windows, window_stats, idx_cfg_typed)
===398: IndexingConfig importable===
112:class FusionConfig(BaseModel):
146:class IndexingConfig(BaseModel):
===397: main.py call sites===
804:     local_video = storage.acquire_local_input(
953:         video_path=req.video_path,
1658:         local_video = storage.acquire_local_input(args.video_path, storage_cfg)
1725:         video_path=args.video_path,
===397: worker correct already===
298:         video_path=local_video,

```

19. assistant (2026-06-29T17:06:59.370Z)

All anchors confirmed exactly as the investigations claim (398: lines 30-35, 43-57, 79-92; 397: main.py:953, 1725, local_video at 804/1658, worker:298). Let me verify the three live-doc link fixes for #399 and confirm the stray-dirs are untracked.

20. user (2026-06-29T17:07:09.823Z)

```

===397: stray dirs untracked?===
git-tracked count above (expect empty)
?? gcs:/
?? gs:/
?? local:/
===399: live doc link targets===
4:**Owner decision required** before any code lands ? see [Open decisions](#open-decisions-for-the-owner).
177:## 4. Open decisions (for the owner)
([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are self-contained.
docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md
target exists
> **Update (2026-06-07):** **AUDIO is now a first-class capture requirement** (new row below),
per ADR-007 ? we will use the shuttle "thwack" as a rally cue (audio+visual fusion lifts recall). This is a hard "record with audio on, mic unobstructed" ask, not a placeholder;
efficacy on real GoPro audio is being validated (experiment E1 in
[AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md)).
docs/archives/research/AUDIO_RALLY_DETECTION_PLAN.md
audio target exists
===399: promotion.py exists===
backend/eval/promotion.py

```

```

38: def served_regresses(
117: def promote_if_served_safe(
===399: leak_scan pattern (tools)===
tools/leak_scan.py
34: # PII / attribution leak guard (tools/leak_scan.py): fails on a committed 32-hex MD5
38: leak-scan:
48:         run: python -m tools.leak_scan

```

21. assistant (2026-06-29T17:08:26.721Z)

All facts confirmed: stray dirs untracked, all three #399 live-doc link targets verified (the `4-` slug fix, the archived PER_VIDEO target exists, the `../archives/research/` AUDIO target exists), `promotion.py` has both functions, and `tools/leak\_scan.py` + a `leak-scan` CI job exist as the precedent for the link-checker. The doc-archival investigation reports zero archive-ready docs (cluster is gated).

I have everything needed. Producing the final merged plan.

Badminton-Highlight-Indexer Wrap-Up ? Ordered Implementation Plan

5 work items, **one PR each**, all branched off `master`, no stacking. Each PR runs the full gate before merge:

...

```
git checkout master && git pull && git checkout -b <branch>
```

...edits...

```
python run_all_tests.py && ruff check . && mypy . && python run_all_tests.py --cov # coverage
must not drop
```

then merge

...

Verified against the tree at `/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer` (on `master`). Line anchors below are confirmed accurate as of this checkout.

Recommended MERGE ORDER (lowest-risk first)

#	PR	Branch	Risk	Why this slot
1	#397	stray-dirs	Lowest	2 one-line edits + .gitignore; identity for local inputs; new test fails-then-passes. Also cleans the repo so later PRs branch clean.
2	#398	served-gate	Low	Eval-helper only; production byte-identical.
3	reportlab-uncap		BLOCKED	DO NOT DO ? see below. No reportlab cap exists in this repo. Skip or convert to a no-op/closed issue.
4	#399	process-hardening	Low-Med	Net-new PR template + CI link-job + 3 live doc-link fixes + 1 cross-link. Touches CI; land after the code fixes are green.
5	doc-archival		N/A	DO NOT ARCHIVE ? Cluster is gated; zero archive-ready docs. Only the 3 small independent fixes below. Last because its edits touch many inbound refs.

If reportlab-uncap is dropped, effective merge order is **#397 ? #398 ? #399 ? doc-archival-housekeeping**.

PR 1 ? #397 Stray scheme-prefixed report dirs · Risk: LOWEST

Root cause (confirmed & reproduced): `obsreport/io.py:48-49` does

``os.path.dirname(os.path.abspath(video_path))``; ``abspath("gcs://b/x.mp4")`` collapses ``///`` and, lacking a leading ``/``, joins under cwd ? ``<repo_root>/gcs://b/x.mp4``, then ``os.makedirs`` creates it. The raw URI leaks in at two callers; the worker already passes the resolved path.

****Files to change**** (all absolute):

- ``/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/main.py``
 - Line ****953**** (API path): ``video_path=req.video_path,`` ? ``video_path=local_video,``
 - Line ****1725**** (CLI path): ``video_path=args.video_path,`` ? ``video_path=local_video,``
 - ``local_video`` is in scope at both (assigned at 804 before the `try@828`; and 1658 before the `try@1662`). Mirrors the already-correct worker (``backend/worker/__main__.py:298``). DB provenance unchanged (``db.add_video`` still stores the source URI).
- ``/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/.gitignore`` ? append (anchored to repo root):
````  
/gcs:/  
/gs:/  
/local:/  
/s3:/  
/gdrive:/  
````

****Tests:**** ADD

``tests/test_api_endpoints.py::test_process_uri_input_writes_no_scheme_dir_at_repo_root`` ? drives ``/api/process`` with ``gs://``, ``gcs://``, ``local://`` through the un-mocked ``emit_indexer_report`` and asserts no scheme dir lands at cwd. Fails on current code, passes after the fix. ****Guard** with ``pytest.importorskip("obsreport")`` for CI portability (writer is a no-op without it).

****Cleanup** (run from repo root, after merge or as the first commit): ****the 3 dirs are untracked** (``git status`` shows ``?? gcs:/ ?? gs:/ ?? local:/``), 48 regenerable report artifacts only:
````

```
rm -rf -- 'gcs:' 'gs:' 'local:'
````
```

****Coverage:**** net ****up**** (new test exercises a previously-untested writer path on the API+CLI branches). ****Do NOT**** edit the sibling ``sports-obsreport`` repo ? the 2-line host fix is sufficient and lower-risk (the `obsreport`` ``scheme://`` guard is a noted-but-out-of-scope option).

PR 2 ? #398 served_gate_a passes a dict where production passes a typed config · Risk: LOW

****Root cause (confirmed):**** ``served_gate_a.served_idx_cfg`` returns a plain dict and feeds it straight into the production seams ``seg._enrich_candidate_stats(...)`` / ``seg._select_for_handoff(...)``, which reach ``fusion_hybrid.py:135`` `fcfg = idx_cfg.fusion`` ? ``AttributeError: 'dict' object has no attribute 'fusion'``. Production always passes a typed ``IndexingConfig`` (via ``_normalize_config`` ? ``AppConfig(**config)``). **\*\*Fix = Option A\*\*** (return a typed config; exercises the real contract + ``extra='forbid'``). Option B (make the seams tolerate dict-or-typed) is ****rejected**** ? it weakens the very contract the litmus exists to test.

****Files to change:****

- ``/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/backend/eval/served_gate_a.py``
 1. Add import after line 33 (``from backend.eval.metrics import Interval, Score, score``): ``from backend.config.models import IndexingConfig``
 2. ``_served_idx_cfg`` (lines 43-57): return signature ``-> IndexingConfig`` and ``return IndexingConfig(fusion={...}, wasb={...}, dead_time_merge_gap=PROD_MERGE_GAP, min_action_window_duration=PROD_MIN_WINDOW)`` (keyword args; nested dicts coerce).
 3. In ``served_handoff_windows`` (lines 79-92): delete the ``idx_cfg_typed: Any = idx_cfg`` # type: ignore` silencer and pass ``idx_cfg`` directly to both seams.
 - Leave the ``Dict`/`Any`` typing imports as-is (still used elsewhere). ****No change** to ``fusion_hybrid.py`` (line 135 is correct against the typed contract).

****Tests:****

- Existing 5 tests in ``tests/test_served_gate_a_regression.py`` become `RUNNABLE+GREEN`; ****no**

assertion edits**. Pinned bands stay valid (re-verified on the 15-clip golden set: served ?F1 ?0.0233, f1_on/f1_off 0.9017, seg_on 0.4696 ? seg_off 0.5825; PROPOSAL ?F1 0.0287). **Do NOT re-pin** ? bands only change when the corpus changes (test docstring already says so).

- **Recommended ADD** (CI-coverable without golden data): a fast synthetic unit test asserting ``_served_idx_cfg(2)`` returns an ``IndexingConfig`` with ``fusion.min_crossings == 2``, and that ``_enrich_candidate_stats`` accepts it without ``AttributeError`` on a tiny synthetic trajectory. The golden-gated tests can't cover this regression in CI; this one can.

****Coverage:**** unchanged in CI by default (the 5 golden tests still skip without a manifest); the optional synthetic test adds real CI coverage of the dict-vs-typed regression. To run the golden suite on the owner box only:

```
...  
SERVED_GATE_A_MANIFEST=/home/avidullu/projects/khelsutra-guru/badminton-highlight-  
indexer/output/p2b_golden/manifest_local.json \  
python -m pytest tests/test_served_gate_a_regression.py -q  
...
```

****Do NOT**** make CI depend on golden data (keeps the ``_golden_specs() >= 6`` gate + ``importorskip("cv2")``). One behavior note: ``extra='forbid'`` now turns a future typo'd served key into a hard error at construction (intended stricter contract).

PR 3 ? reportlab-uncap · Risk: BLOCKED ? DO NOT IMPLEMENT AS SPECIFIED

****Flag:**** This work item has ****no investigation result in the provided JSON**** and ****no verifiable target in the tree****. I searched exhaustively and found ****no reportlab version cap anywhere****:

- Not in ``requirements.txt`` (reportlab is not even listed) or ``pyproject.toml`` ``[project.dependencies]`` / ``optional-dependencies``.
- Not in ``.github/workflows/*.yaml``, ``deploy/**``, any ``constraints*.txt`` (none exist), or the sibling ``sports-obsreport/pyproject.toml``.
- The only references are ``backend/eval/ablation_pdf.py`` (lazy ``from reportlab.lib import ...``, unversioned) and two ``pytest.importorskip("reportlab")`` guards in ``tests/test_ablation.py``. reportlab is an ****undeclared, lazily-imported soft dependency**** ? there is nothing to "uncap."

****Recommended action:**** ****do not open a code PR.**** Take ONE of:

1. ****Close**** the item as not-applicable (no cap exists), OR
2. If the real intent is to ***declare*** reportlab as an optional extra (it's currently undeclared, which is the actual latent issue), do that explicitly instead ? add a ``pdf = ["reportlab>=4,<5"]`` optional group in ``pyproject.toml`` with a ``tests/test_ablation.py`` guard already present. ****Confirm intent with the owner before coding**** ? declaring-with-a-cap is the opposite of "uncap," so the item title likely refers to a constraint that lives outside this repo.

Either way, ****block this PR until the target is confirmed.**** Do not fabricate a requirements edit.

PR 4 ? #399 Launch-PR process hardening · Risk: LOW-MED

Four coordinated parts. ****Scope decision:** fix the 3 LIVE broken links only; treat the 49 ``docs/archives/**`` links as historical-by-design (DOC_STATUS §2 step 5) and exclude ``docs/archives/**`` from the committed checker. ****Do NOT mass-rewrite the 49 archived `../` links.**

****Files to change:****

- ``/home/avidullu/.../docs/I18N_PLAN.md`` ? line 4 anchor: ``(#open-decisions-for-the-owner)`` ? ``(#4-open-decisions-for-the-owner)`` (heading ``## 4. Open decisions (for the owner)`` slugs to ``4-open-decisions-for-the-owner`` ? ****verified****).
- ``/home/avidullu/.../docs/QUALITY_ITERATIONS.md`` ? line 621: ``(PER_VIDEO_OUTPUT_LAYOUT.md)`` ? ``(archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md)`` (****target exists****).
- ``/home/avidullu/.../docs/data_pipeline/VIDEO_RECORDING_GUIDELINES.md`` ? line 10: ``(archives/research/AUDIO_RALLY_DETECTION_PLAN.md)`` ?

```
`(./archives/research/AUDIO_RALLY_DETECTION_PLAN.md)` (doc lives one level deep; **target
exists at `docs/archives/research/...`**).
- `/home/avidullu/.../docs/RALLY_DETECTION_FIXES_PLAN.md` ? §8 (line ~110-111): make the
"(promote iff tolF1 does not regress)" rule a link to `../backend/eval/promotion.py` (file
exists; `served_regresses` @38, `promote_if_served_safe` @117 ? **verified**). The phrase wraps
across 110-111; fold the link text across the wrap.
- `/home/avidullu/.../CLAUDE.md` ? add one bullet to "Workflow conventions" (after the
PROJECT_DOC_TEMPLATE bullet, ~line 68): a launch/bit-flip PR MUST run the DOC_STATUS §2 archival
checklist in the same PR, enforced by the new PR template.
- `/home/avidullu/.../.github/PULL_REQUEST_TEMPLATE.md` ? **NEW** (none exists): the "Launch /
bit-flip PRs ? archival gate" checklist block (verify-vs-code ? flip status ? update inbound
refs ? keep lineage ? `git mv` to archives ? record in §3).
- `/home/avidullu/.../tools/check_md_links.py` ? **NEW**, stdlib-only, mirrors the existing
`tools/leak_scan.py` (run as `python -m tools.check_md_links`); **excludes `docs/archives/`**.
- `/home/avidullu/.../.github/workflows/ci.yml` ? add a `link-check` job (sibling of the
existing `leak-scan` job @38, same `python -m tools.X` shape, no pip install).
```

****Tests:**** ADD (recommended) `tests/test\_check\_md\_links.py` ? tmp_path fixtures: missing path
flagged, dead anchor flagged, valid link passes, http/mailto/fenced-code skipped,
`docs/archives/` excluded.

****Coverage:**** net up (new tool + its test). The CI link-check job fails only on ****live****-doc
breakage.

****Do NOT:**** mass-rewrite the 49 archived links; cover `backend/deploy/scripts` READMEs (out of
the issue's root+`docs/` scope); use a multi-template `.github/PULL\_REQUEST\_TEMPLATE/` directory
(single-file form is correct here).

PR 5 ? Doc-archival · Risk: N/A ? DO NOT ARCHIVE ANY DOC NOW

****Hard finding:** `archive\_ready\_docs = []`. Zero docs are archive-ready. ****** The whole
R-series/quality cluster is ****ACTIVE and gated**** (DOC_STATUS §4 finding #3). P2b (#396) promoted
the operational `inactivity\_temporal\_density` flag (do-no-harm, ?tolF1 +0.010, CI spans 0,
p=0.18) ? it is ****NOT**** the R2 tolF1 metric-gate ablation and ****NOT**** the §7.1 single-number
attribution, so it does ****NOT**** close the gate.

****Explicitly DO NOT archive / DO NOT `git mv`:****

- `RALLY\_DETECTION\_FIXES\_PLAN.md`, `RALLY\_DETECTION\_CODE\_REVIEW.md` ? DONE in content but their
own headers defer archival to the cluster gate. Keep in place.
- `RALLY\_QUALITY\_RESEARCH.md`, `R2\_EVAL\_METRIC\_DESIGN.md`, `RALLY\_DETECTION\_GAP\_CLOSURE.md`,
`PROMINENT\_COURT\_DETECTION.md`, `REAL\_FOOTAGE\_VALIDATION\_REPORT.md`,
`RALLY\_DETECTION\_QUALITY\_REPORT.md`, `R1\_MILESTONE\_LAUNCH\_REPORT.md` ? DoD not met / part of the
gated cluster.
- `QUALITY\_ITERATIONS.md`, `HOW\_RALLY\_DETECTION\_WORKS.md` ? ****LIVING****, "Keep" per DOC_STATUS
§3b/§3c ? ****never**** archive as a whole.

When the gate eventually clears, archive the ****whole cluster as one unit**** via DOC_STATUS §2
steps 1-6 (verify ? flip status ? update inbound refs `README.md:1/2/4/7` + `NEXT\_STEPS.md`
items 8/9/9b + DOC_STATUS §3 rows ? keep lineage ? `git mv` to `docs/archives/past\_projects/` ?
record). ****Not now.****

****This PR's actual scope = 3 small independent housekeeping fixes**** (do NOT touch any doc's
location):

1. `docs/DOC\_STATUS.md` §3 ? add tracking rows for `RALLY\_DETECTION\_FIXES\_PLAN.md` (DONE /
gated-to-R-cluster) and `RALLY\_DETECTION\_CODE\_REVIEW.md` (terminal snapshot / gated) ? currently
untracked.
2. `docs/NEXT\_STEPS.md` line 25 (item 9b) ? STALE: still says PR #393 must do the A/B flip "only
if tolF1 doesn't regress," but #396 already did it and promoted default-ON. Update/remove 9b;
re-check item 8 @ line 21.
3. `docs/RALLY\_DETECTION\_FIXES\_PLAN.md` header ? STALE self-claim that it is "indexed in
docs/README.md" (grep of README finds no entry). Remove/correct.

****Files to change:**** `docs/DOC\_STATUS.md`, `docs/NEXT\_STEPS.md`,
`docs/RALLY\_DETECTION\_FIXES\_PLAN.md`. ****Tests:**** none (docs only); gate is ruff/mypy/tests pass
+ coverage unchanged. ****Last in the order**** ? these edits touch the doc cross-ref graph, so land
them after everything else is green.

Cross-cutting flags

- ****reportlab-uncap** is unverifiable in this repo ? block/clarify before any code (PR 3).******
- ****No doc is archive-ready ? PR 5 must NOT `git mv` anything; it is 3 staleness fixes only.****
- For #399, fix the ****3 live**** links only; the ****49 archived**** links are historical-by-design ?
exclude `docs/archives/\*\*` from the committed checker rather than rewriting them.
- For #398, do ****not**** re-pin bands and do ****not**** make CI depend on golden data.
- For #397, do ****not**** edit the sibling `sports-obsreport` repo; the 2-line host fix suffices.

Plan source files referenced are all under `/home/avidullu/projects/khelsutra-guru/badminton-
highlight-indexer/` (and the sibling `/home/avidullu/projects/khelsutra-guru/sports-
obsreport/src/obsreport/io.py:48-49` as the documented root-cause writer for #397, left
unmodified).